Research Highlights

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- A Detection Quality Index (DQI) fusing eight vibration-based indicators with exponential false-positive penalisation improves damage detection reliability in noisy offshore environments.
- Secondary elements (braces) behave as structural fuses, detectable from 30% severity, achieving 92.2% (denting) and 96.5% (corrosion) global detection rates.
- Main elements (legs) require damage severity exceeding 50% for reliable detection, indicating strong structural redundancy.
- A Topology-Weighted Precision metric corrects classical Precision underestimation by up to +22 percentage points for brace elements in jacket lattice structures.
- Validated over 3,240 simulation runs on a calibrated finite-element model under corrosion and denting scenarios, with AUC-ROC > 0.90 for brace and inclined-leg members.